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Comparative study of human and mouse pregnane X receptor agonistic activity in 200 pesticides using *in vitro* reporter gene assays

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ABSTRACT

The nuclear receptor, pregnane X receptor (PXR), is a ligand-dependent transcription factor that regulates genes involved in xenobiotic metabolism. Recent studies have shown that PXR activation may affect energy metabolism as well as the endocrine and immune systems. In this study, we characterized and compared the agonistic activities of a variety of pesticides against human PXR (hPXR) and mouse PXR (mPXR). We tested the hPXR and mPXR agonistic activity of 200 pesticides (29 organochlorines, 11 diphenyl ethers, 56 organophosphorus pesticides, 12 pyrethroids, 22 carbamates, 12 acid amides, 7 triazines, 7 ureas, and 44 others) by reporter gene assays using COS-7 simian kidney cells. Of the 200 pesticides tested, 106 and 93 activated hPXR and mPXR, respectively, and a total of 111 had hPXR and/or mPXR agonistic activity with greater or lesser inter-species differences. Although all of the pyrethroids and most of the organochlorines and acid amides acted as PXR agonists, a wide range of pesticides with diverse structures also showed hPXR and/or mPXR agonistic activity. Among the 200 pesticides, pyributicarb, pretilachlor, piperophos and butamifos for hPXR, and phosalone, prochloraz, pendimethalin, and butamifos for mPXR, acted as particularly potent activators at low concentrations in the order of 10^{-8}–10^{-7} M. In addition, we found that several organophosphorus oxon- and pyributicarb oxon-metabolites decreased PXR activation potency compared to their parent compounds. These results suggest that a large number of structurally diverse pesticides and their metabolites possess PXR-mediated transcriptional activity, and their ability to do so varies in a species-dependent manner in humans and mice.

1. Introduction

The pregnane X receptor (PXR, NR1I2) is a member of the nuclear receptor superfamily of ligand-activated transcription factors, and is highly expressed in the liver, small intestine, and colon in the human, rabbit, rat, and mouse (Kliewer et al., 2002; Xie and Evans, 2001). PXR is activated by a variety of structurally diverse array of endogenous and exogenous chemicals including steroids, antibiotics, antimycotics, bile acids (Staudinger et al., 2001; Xie et al., 2001), and environmental pollutants, such as pesticides (Lemaire et al., 2004; Matsubara et al., 2007; Mikamo et al., 2003) and flame retardants (Honkakoski et al., 2004; Pacyniak et al., 2007), since its enlarged ligand binding pocket is flexible enough to accommodate these diverse chemicals (Chrencik et al., 2005). On interaction with these ligands, PXR forms a heterodimer with the retinoid X receptor (RXR, NR2B), another member of the nuclear receptor superfamily, in the nucleus and binds to a xenobiotic response element (XRE) located in the promoter region of the target gene, thereby regulating its transcription (Goodwin et al., 1999). Such target genes containing XRE motifs include microsomal cytochrome P450 (CYP) isoforms, such as CYP3A4, CYP2B6 and CYP2C8, conjugation enzymes, such as UDP-glucuronosyltransferase (UGT1A1) and sulfotransferase, and ABC family transporters, such as MDR1 and MRP2 (Kliewer et al., 2002; Synold et al., 2001). Although the induced proteins are responsible for the metabolism, deactivation, and transport of environmental chemicals and drugs, they are also involved in bile acid, thyroid, and steroid hormone catabolism, so that xenobiotics may interfere with normal physiological functions. Moreover, recent studies have provided evidence that PXR activation is involved in lipid metabolism, glucose homeostasis, and inflammation (Moreau et al., 2008; Zhou et al., 2009).

Structurally, PXR is similar to other nuclear receptors, and includes an N-terminal transactivating domain, followed by a DNA binding domain (DBD), and a C-terminal ligand-binding domain (LBD). Cloning and characterization of PXR from humans, rabbits, rats and mice has shown that there is >95% sequence identity in the DBD regions. However, the LBDs of the PXRs show lower identity and share only 70-80% amino acid identity; e.g., the percent

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sequence identity in the LBD between human PXR (hPXR) and mouse PXR (mPXR) is 77.3% (Jones et al., 2000). Therefore, species differences in CYP3A regulation by xenobiotics are likely due to differences in the LBD, as the DBD is highly conserved among the PXRs from various species. The antibiotic rifampicin (RIF), for example, is an activator of both hPXR and rabbit PXR, but has little activity on mPXR or rat PXR (rPXR) (Jones et al., 2000). In contrast, pregnenolone 16-α-carbonitrile (PCN) is a more potent activator of mPXR and rPXR than of hPXR or rabbit PXR (Jones et al., 2000).

The ubiquitous nature of pesticide usage with minimal precautions has resulted in the contamination of food, the workplace, and the natural environment in general. Pesticides possess a wide variety of chemical structures and may represent the most likely candidates for ligands against nuclear receptors. Previous studies using *in vitro* assays reported that several pesticides, including organochlorines and pyrethroids, possess hPXR agonistic activity (Coumoul et al., 2002; Lemaire et al., 2004, 2006). In addition, using HepG2-derived cells stably expressing the CYP3A4 promoter/enhancer, Matsubara et al. (2007) also demonstrated that a variety of pesticides induce CYP3A4 via hPXR. Furthermore, Mikamo et al. (2003) reported that some pesticides, including methoxychlor, activated rPXR in yeast two hybrid assays, and induced CYP3A1, CYP2B1, and CYP2C11 in the rat liver after intraperitoneal injection. Taken together, although several pesticides have human and rodent PXR agonistic activity, it is not fully understood whether numerous other pesticides also activate PXR, or whether their activities are dependent on species-specific differences between humans and rodents.

We have previously investigated 200 pesticides, including related isomers and metabolites, for estrogenicity via human estrogen receptor (hER) subtypes, hERα and hERβ, and androgenicity via a human androgen receptor (hAR) by *in vitro* reporter gene assays (Kojima et al., 2004). Consequently, 47, 33 and 66 of 200 pesticides were shown to act as hERα and hERβ agonists and hAR antagonists, respectively. Furthermore, our recent papers have reported that 3 and 12 of these 200 pesticides possess *in vitro* peroxisome proliferator-activated receptor (PPAR)α and aryl hydrocarbon receptor (AhR) agonistic activity, respectively (Takeuchi et al., 2006, 2008). In this study, we examined the hPXR and mPXR agonistic activity of the 200 above-mentioned pesticides *in vitro* by reporter gene assays using COS-7 simian kidney cells, and found that 111 of the 200 pesticides, including many newly identified pesticides, were hPXR and/or mPXR activators with greater or lesser inter-species differences. Although the structures of these PXR-activating pesticides are chemically diverse, we identified certain structural aspects that appear to be necessary for activation. Thus, we here provide evidence that a large number of pesticides act as hPXR and mPXR agonists.

2. Materials and methods

2.1. Chemicals

The 200 pesticides were classified into nine groups of 29 organochlorines, 11 diphenyl ethers, 56 organophosphorus pesticides, 12 pyrethroids, 22 carbamates, 12 acid amides, 7 triazines, 7 ureas, and 44 others (Table 1). These pesticides had a purity of 97–100% and were purchased from Wako Pure Chemical Industries Ltd. (Wako, Osaka, Japan), Sigma–Aldrich (St. Louis, MO, USA), Dr. Ehrenstorfer GmbH (Augsburg, Germany), AccuStandard Inc. (New Haven, CT, USA), and Hayashi Pure Chemical Industries Ltd. (Hayashi; Osaka, Japan). We purchased butamifos oxon (99% pure), EPN oxon (>97% pure), isofenphos oxon (99% pure), isoxathion oxon (>97% pure), and piperophos oxon (>98% pure) from Wako, and chlorpyrifos oxon (99.9% pure) from Hayashi. We synthesized pyributicarb oxon (99% pure) according to the method previously reported by Kanetoshi et al. (1999). We purchased two typical PXR ligands, RIF (>97% pure) and PCN (>98% pure), from Sigma–Aldrich, and used them as positive controls in the hPXR and mPXR assays, respectively. We used DMSO (Wako) as a vehicle, and all compounds were dissolved in DMSO at a concentration of 1×10^{-2} M.

Table 1
Two hundred pesticides tested in the reporter gene assays for hPXR and mPXR.

Pesticide name	CAS No.
1 Organochlorines (29)	
Aldrin	309-002
α-BHC	319-84-6
β-BHC	319-85-7
γ-BHC	58-89-9
δ-BHC	319-86-8
Captan	133-06-2
cis-Chlordane	5103-74-2
trans-Chlordane	5103-71-9
Chlorobenzilate	510-15-6
Chloropropylate	5836-10-2
Chlorothalonil	1897-45-6
o,p′-DDT	789-02-6
p,p′-DDT	50-29-3
p,p′-DDE	72-55-9
p,p′-DDD	72-54-8
Dichlobenil	1194-65-6
Dicofol	115-32-2
Dieldrin	60-57-1
α-Endosulfan	959-98-8
β-Endosulfan	33213-65-9
Endosulfan sulfate	1031-07-8
Endrin	72-20-8
Folpet	133-07-3
Fthalide	27355-22-2
Heptachlor	76-44-8
Heptachlor epoxide	1024-57-3
Methoxychlor	72-43-5
Pentachlorophenol	87-86-5
Quintozene	82-68-6
2 Diphenyl ethers (11)	
Acifluorfen	50594-66-6
Acifluorfen-methyl	50594-67-7
Bifenox	42576-02-3
Chlomethoxynil	32861-85-1
Chlornitrofen	1836-77-7
Chlornitrofen-amino	26306-61-6
Chloroxurone	1982-47-4
Dichlofop-methyl	51338-27-3
Fluazifop-butyl	69806-50-4
Nitrofen	1836-75-5
Oxyfluorfen	42874-03-3
3 Organophosphorus (56)	
Acephate	30560-19-1
Anilofos	64249-01-0
Bromophos-ethyl	4824-78-6
Bromophos-methyl	2104-96-3
Butamifos	36335-67-8
Chlorpyrifos	2921-88-2
Chlorpyrifos-methyl	5598-13-0
Cyanofenphos	13067-93-1
Cyanophos	2636-26-2
Diazinon	333-41-5
Dichlorfenthion	97-17-6
Dichlorvos	62-73-7
Dimethoate	60-51-5
Dioxabenzofos	3811-49-2
Disulfoton	298-04-4
EPN[a]	2104-64-5
Edifenphos	17109-49-8
Ethion	563-12-2
Ethoprophos	13194-48-4
Fenamiphos	22224-92-6
Fenchlorphos	299-84-3
Fenitrothion	122-14-5
Fenitrothion oxon	2255-17-6
Fensulfothion	115-90-2
Fenthion	55-38-9
Glyphosate	1071-83-6
Iprobenfos	26087-47-8
Isofenphos	25311-71-1
Isoxathion	18854-01-8
Leptophos	21609-90-5
Malathion	121-75-5

Table 1 (*Continued*)

Pesticide name	CAS No.
Mecarbam	2595-54-2
Methamidophos	10265-92-6
Methidathion	950-37-8
Methyl-parathion	298-00-0
Monocrotophos	6923-22-4
Parathion	56-38-2
Phenthoate	2597-03-7
Phorate	298-02-2
Phosalone	2310-17-0
Phosmet	732-11-6
Piperophos	24151-93-7
Pirimiphos-methyl	29232-93-7
Profenofos	41198-08-7
Propaphos	7292-16-2
Prothiofos	34643-46-4
Prothiofos oxon	38527-91-2
Pyridaphenthion	119-12-0
Quinalphos	13593-03-8
Terbufos	13071-79-9
Tetrachlorvinphos	22248-79-9
Thiometon	640-15-3
Tolclofos-methyl	57018-04-9
Tolclofos-methyl oxon	97483-08-4
Trichlorfon	52-68-6
Vamidothion	2275-23-2
4 Pyrethroids (12)	
Cyfluthrin	68359-37-5
Cyhalothrin	68085-85-8
Cypermethrin	52315-07-8
Deltamethrin	52918-63-5
Etofenprox	80844-07-1
Fenvalerate	51630-58-1
Flucythrinate	70124-77-5
Fluvalinate	69409-94-5
Permethrin	52645-53-1
Pyrethrin	8003-34-7
Tefluthrin	79538-32-2
Tralomethrin	66841-25-6
5 Carbamates (22)	
Bendiocarb	22781-23-2
Benomyl	17804-35-2
Carbaryl	63-25-2
Carbendazim	10605-21-7
Carbofuran	1563-66-2
Chlorpropham	101-21-3
Diethofencarb	87130-20-9
Dimepiperate	61432-55-1
Esprocarb	85785-20-2
Ethiofencarb	29973-13-5
Fenobucarb	3766-81-2
Isoprocarb	2631-40-5
Methiocarb	2032-65-7
Methomyl	16752-77-5
Molinate	2212-67-1
Oxamyl	23135-22-0
Phenmedipham	13684-63-4
Pirimicarb	23103-98-2
Pyributicarb	88678-67-5
Thiobencarb	28249-77-6
Thiobencarb sulfon	none
Thiram	137-26-8
6 Acid amides (12)	
Alachlor	15972-60-8
Asulam	3337-71-1
Cafenstrole	125306-83-4
Flutolanil	66332-96-5
Mefenacet	73250-68-7
Mepronil	55814-41-0
Metalaxyl	57837-19-1
Metolachlor	51218-45-2
Pretilachlor	51218-49-6
Propanil	709-98-8
Propyzamide	23950-58-5
Thenylchlor	96491-05-3

Table 1 (*Continued*)

Pesticide name	CAS No.
7 Triazines (7)	
Anilazine	101-05-3
Atrazine	1912-24-9
Metribuzin	21087-64-9
Prometon	1610-18-0
Prometryn	7287-19-6
Simazine	122-34-9
Simetryn	1014-70-6
8 Ureas (7)	
Bensulfuron-methyl	83055-99-6
Dymuron	42609-52-9
Diflubenzuron	35367-38-5
Diuron	330-54-1
Linuron	330-55-2
Pencycuron	66063-05-6
Prochloraz	67747-09-5
9 Others (44)	
Amitraz	33089-61-1
Benfuresate	68505-69-1
Bentazone	25057-89-0
Benzoximate	29104-30-1
Biphenyl	92-52-4
Bitertanol	55179-31-2
Bromopropylate	18181-80-1
Chinomethionat	2439-01-2
Chloridazon (PAC)	1698-60-8
Dazomet	533-74-4
Diquat	2764-72-9
Ethoxyquin	91-53-2
Fenarimol	60168-88-9
Ferimzone	89269-64-7
Fluazinam	79622-59-6
Imazalil	35554-44-0
Imidacloprid	138261-41-3
Iminoctadine	13516-27-3
Indanofan	133220-30-1
Ioxynil	3861-47-0
Iprodione	36734-19-7
Isoprothiolane	50512-35-1
Lenacil	2164-08-1
MCPA[b]	94-74-6
2,4-D[c]	94-75-7
Paraquat	1910-42-5
Pendimethalin	40487-42-1
2-Phenylphenol	90-43-7
Probenazole	27605-76-1
Procymidone	32809-16-8
Propiconazole	60207-90-1
Pyrazolynate	58011-68-0
Pyrazoxyfen	71561-11-0
Pyroquilon	57369-32-1
Sethoxydim	74051-80-2
Thiabendazole	148-79-8
Thiocyclam	31895-22-4
Thiophanate-methyl	23564-05-8
Triadimefon	43121-43-3
Tricyclazole	41814-78-2
Triflumizole	68694-11-1
Trifluralin	1582-09-8
Triforine	26644-46-2
Vinclozolin	50471-44-8

[a] O-ethyl-O-p-nitrophenyl phenylphosphonothioate.
[b] 4-Chloro-o-toloxyacetic acid.
[c] 2,4-Dichlorophenoxyacetic acid.

2.2. Cell line and cell culture materials

We obtained COS-7 simian kidney cells from the American Type Culture Collection, fetal bovine serum (FBS) and charcoal-dextran treated FBS (CD-FBS) from Hyclone (Logan, UT, USA), Dulbecco's modified Eagle's medium (DMEM) from GIBCO-BRL (Invitrogen, Rockville, MD, USA), glutamine solution and penicillin–streptomycin (antibiotics) solution from Dainippon Pharmaceutical Co. Ltd. (Osaka, Japan), and 0.25% trypsin/0.02% ethylenediamine tetra-acetic acid (EDTA) disodium salt solution from Life Technologies (Paisley, UK).

2.3. Plasmids

The expression plasmids of pSG5-hPXR and pSG5-mPXR (Lehmann et al., 1998) encoding the full-length receptor protein were kindly provided by Steven Kliewer (Department of Molecular Biology, University of Texas, Southwestern Medical Center, Dallas, TX, USA), and the reporter plasmid pXREM-3A4-Luc (Goodwin et al., 1999) was kindly provided by Bryan Goodwin (High Throughput Biology, Discovery Research, GlaxoSmithKline, Research Triangle Park, NC, USA). We purchased the internal control plasmid, pCMVβ-Gal, from Clontech (Palo Alto, CA, USA).

2.4. Transfection of plasmids to cells and luciferase activity assay

We plated the host COS-7 cells in 96-well microtiter plates (Nalge, Nunc, Denmark) at a density of 8400 cells per well in DMEM containing 10% CD-FBS (complete medium) 1 day before transfection. For detection of hPXR or mPXR activity, cells were transfected with 12 ng pSG5-hPXR or pSG5-mPXR, 48 ng pXREM-3A4-Luc, and 12 ng pCMVβ-Gal per well using FuGENE®6 Transfection Reagent (Roche Diagnostics Corp., Indianapolis, IN, USA). After a 3-h transfection period, cells were dosed with various concentrations of the test compounds or with 0.1% DMSO (vehicle control) in complete medium. To avoid any cytotoxic effects associated with the test compounds, we performed screening assays for test compounds at concentrations from 1×10^{-7} to 1×10^{-5} M, except for chlorothalonil and fluazinam, which were screened at concentrations from 1×10^{-8} to 1×10^{-6} M. After an incubation period of 24 h, cells were rinsed with phosphate-buffered saline (pH 7.4) and lysed with passive lysis buffer (50 μl/well; Promega, Madison, WI, USA). In addition, to confirm that the luciferase inductions of test chemicals are PXR-dependent, we also performed another assay using COS-7 cells transfected with 48 ng pXREM-3A4-Luc, and 12 ng pCMVβ-Gal per well (without the PXR expression plasmid).

We measured the *firefly* luciferase activity in a 5-μl aliquot of the cell lysate in one reaction tube with a MiniLumat LB 9506 luminometer (Berthold, Wildbad, Germany) using the Luciferase Assay System (Promega), according to the manufacturer's instructions. The luciferase activity was normalized against the β-galactosidase activity for each treatment. Results are expressed as mean ± SD from at least three independent experiments performed in triplicate.

2.5. β-Galactosidase activity assay

We obtained bovine serum albumin (BSA) and 4-methylumbelliferyl-β-D-galactoside (4-MUG) from Sigma–Aldrich. We measured β-galactosidase activity to check the cytotoxicity of the test chemicals against transcriptional activity using a fluorescence method as described previously (Takeuchi et al., 2005, 2006).

2.6. Evaluation of PXR agonistic activities

To estimate the potency of the receptor agonistic activity of the tested compounds, we presented the luminescence intensity of the assay as a dose–response curve. We obtained the concentration of the compound equal to 20% of the maximal response of positive control (RIF or PCN) from the dose–response curve of the luminescence intensity and expressed it as an REC_{20} value (20% relative effective concentration). When the agonistic activity of the test compound was higher than the REC_{20} value for the concentration tested ($\leq 1 \times 10^{-5}$ M), we judged the pesticides to be positive for agonistic activity against hPXR or mPXR. Each REC_{20} value represents the mean of three independent experiments.

3. Results

3.1. Species-specific activity of typical PXR ligands, RIF and PCN, in the hPXR and mPXR assays

Figure 1A shows the dose–response transactivation of hPXR by RIF and PCN, indicating that hPXR is preferentially activated at very low concentrations of RIF, but not activated even at 1×10^{-5} M of PCN. The maximal hPXR activity of RIF was 8.5-fold that of the vehicle control at 1×10^{-5} M. In contrast, Fig. 1B shows the dose–response transactivation of mPXR by RIF and PCN, indicating that mPXR is preferentially activated at very low concentrations of PCN, but not activated even at 1×10^{-5} M of RIF. Again, the maximal mPXR activity of PCN was 8.5-fold that of the vehicle control at 1×10^{-5} M. From these dose–response curves, we estimated the REC_{20} values of RIF for hPXR and PCN for mPXR to be 4.3×10^{-7} M and 5.7×10^{-8} M, respectively.

3.2. Transcriptional activity of 200 pesticides in the hPXR and mPXR assays

We screened the hPXR and mPXR agonistic activities of the 200 pesticides at concentrations of $\leq 1 \times 10^{-5}$ M using the above transactivation assays. The REC_{20} values and relative luciferase activities (RLA) of 111 pesticides evaluated as having an agonistic effect for hPXR and/or mPXR are summarized in Table 2. In this screening study, we found that 106 pesticides activated hPXR-mediated transcription by more than 20% of the maximum activity of RIF. In addition, we found that 93 pesticides activated mPXR-mediated transcription by more than 20% of the maximum activity of PCN. A total of 111 pesticides possessed hPXR and/or mPXR agonistic activity, and 88 of these pesticides acted as both hPXR and mPXR agonists.

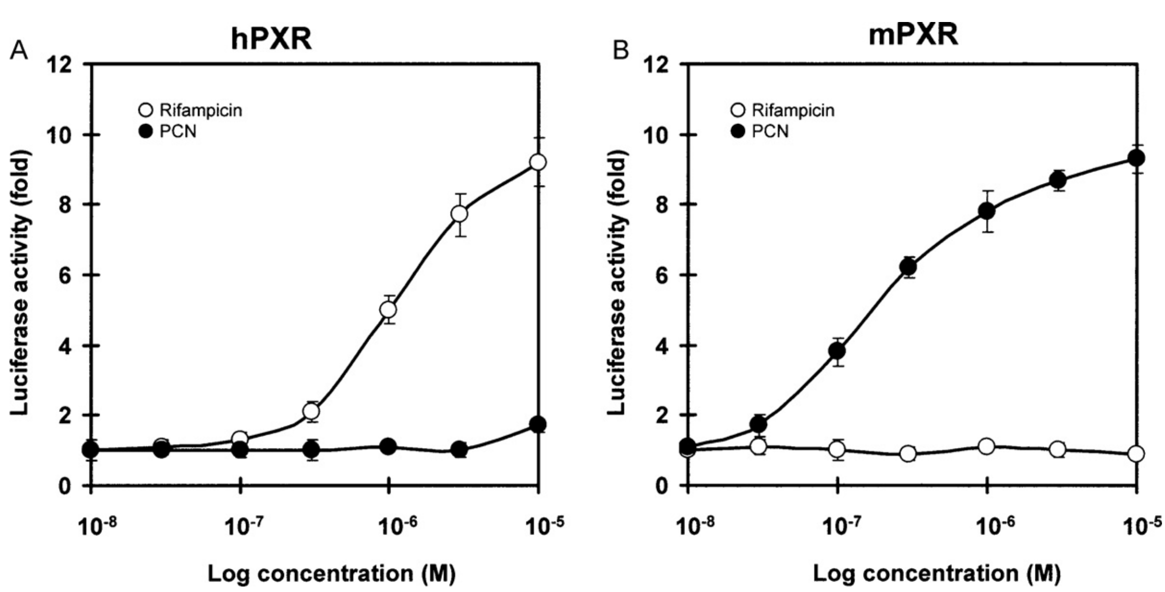

Fig. 1. Species-specific response curves of typical PXR ligands in the hPXR and mPXR transactivation assays. COS-7 cells were transiently transfected with an expression plasmid for human or mouse PXR as well as a reporter-responsive *firefly* luciferase plasmid and a constitutively active β-galactosidase expression plasmid. Cells were treated with increasing concentrations of typical PXR ligands, rifampicin (RIF) and pregnenolone 16α-carbonitrile (PCN). The *firefly* luciferase activity was normalized against β-galactosidase activity. Values represent the means ± SD of three independent experiments and are presented as the mean *n*-fold induction over the vehicle control.

Table 2
Responses induced by pesticides testing positive in the human and/or mouse PXR transactivation assays.

Compound	hPXR assay		mPXR assay	
	REC_{20}[a]	RLA[b]	REC_{20}[a]	RLA[b]
	(M)	(%)	(M)	(%)
Rifampicin	4.3×10^{-7}	100[c]	N.D.[d]	
PCN	N.D.		5.7×10^{-8}	100[e]
1. Organochlorines				
γ-BHC	1.1×10^{-6}	120 ± 13	5.2×10^{-6}	35 ± 1
Chloropropylate	1.3×10^{-6}	132 ± 13	3.3×10^{-6}	38 ± 4
Methoxychlor	1.4×10^{-6}	95 ± 8	1.8×10^{-6}	88 ± 5
α-BHC	1.6×10^{-6}	74 ± 2	6.4×10^{-6}	28 ± 3
β-BHC	2.1×10^{-6}	55 ± 2	N.D.	2 ± 4
trans-Chlordane.	2.6×10^{-6}	48 ± 9	2.8×10^{-6}	33 ± 1
α-Endosulfan	2.6×10^{-6}	53 ± 8	5.8×10^{-6}	33 ± 7
β-Endosulfan	2.8×10^{-6}	56 ± 10	4.5×10^{-6}	23 ± 3
p,p'-DDE	3.0×10^{-6}	40 ± 4	N.D.	16 ± 3
δ-BHC	3.1×10^{-6}	83 ± 5	N.D.	17 ± 3
Heptachlor epoxide	3.4×10^{-6}	50 ± 7	4.6×10^{-6}	40 ± 6
p,p'-DDD	3.7×10^{-6}	34 ± 1	6.0×10^{-6}	33 ± 11
Dicofol	3.8×10^{-6}	58 ± 16	1.6×10^{-6}	105 ± 12
Chlorobenzilate	4.0×10^{-6}	61 ± 5	6.7×10^{-6}	30 ± 1
Dieldrin	4.2×10^{-6}	42 ± 3	2.8×10^{-6}	28 ± 3
o,p'-DDT	4.4×10^{-6}	53 ± 3	2.6×10^{-6}	71 ± 1
Endosulfan sulfate	4.7×10^{-6}	33 ± 5	7.5×10^{-6}	22 ± 1
p,p'-DDT	4.8×10^{-6}	44 ± 6	3.3×10^{-6}	71 ± 2
Heptachlor	5.1×10^{-6}	33 ± 9	8.0×10^{-6}	24 ± 1
Aldrin	5.2×10^{-6}	35 ± 1	N.D.	1 ± 1
Endrin	5.6×10^{-6}	28 ± 7	N.D.	19 ± 1
cis-Chlordane	6.2×10^{-6}	31 ± 1	3.0×10^{-6}	29 ± 1
Quintozene	6.7×10^{-6}	36 ± 9	N.D.	17 ± 1
Fthalide	N.D.	17 ± 1	2.8×10^{-6}	53 ± 7
2. Diphenyl ethers				
Bifenox	2.9×10^{-6}	59 ± 1	3.5×10^{-6}	46 ± 2
Acifluorfen-methyl	3.3×10^{-6}	36 ± 4	4.4×10^{-6}	39 ± 1
Chlornitrofen	3.9×10^{-6}	51 ± 1	3.3×10^{-6}	52 ± 2
Oxyfluorfen	5.0×10^{-6}	30 ± 3	3.7×10^{-6}	34 ± 2
Chlomethoxynil	5.9×10^{-6}	35 ± 4	2.7×10^{-6}	56 ± 4
Nitrofen	9.3×10^{-6}	21 ± 2	7.5×10^{-6}	27 ± 1
3. Organophosphorus				
Piperophos	1.7×10^{-7}	159 ± 16	1.0×10^{-6}	62 ± 5
Butamifos	3.7×10^{-7}	129 ± 10	2.3×10^{-7}	85 ± 16
Isofenphos	6.4×10^{-7}	167 ± 4	6.8×10^{-7}	110 ± 8
Ethion	7.0×10^{-7}	88 ± 1	8.2×10^{-7}	75 ± 9
Anilofos	1.1×10^{-6}	71 ± 1	9.5×10^{-6}	21 ± 3
Phosalone	1.5×10^{-6}	76 ± 6	2.9×10^{-8}	85 ± 3
Iprobenfos	1.5×10^{-6}	74 ± 2	2.2×10^{-6}	57 ± 7
EPN	1.8×10^{-6}	83 ± 5	1.7×10^{-6}	89 ± 3
Fenamiphos	1.8×10^{-6}	75 ± 2	5.2×10^{-6}	38 ± 3
Isoxathion	2.0×10^{-6}	52 ± 5	5.1×10^{-6}	36 ± 1
Cyanofenphos	2.3×10^{-6}	70 ± 4	3.6×10^{-6}	40 ± 7
Phenthoate	3.0×10^{-6}	63 ± 7	5.3×10^{-6}	35 ± 4
Tetrachlorvinphos	3.5×10^{-6}	41 ± 7	2.8×10^{-6}	53 ± 4
Propaphos	3.6×10^{-6}	55 ± 6	2.2×10^{-6}	65 ± 7
Prothiofos	3.6×10^{-6}	40 ± 6	3.8×10^{-6}	53 ± 3
Pyridaphenthion	4.4×10^{-6}	33 ± 1	1.2×10^{-6}	80 ± 3
Bromophos-ethyl	4.6×10^{-6}	70 ± 12	2.7×10^{-6}	46 ± 1
Chlorpyrifos	4.7×10^{-6}	35 ± 3	4.5×10^{-6}	44 ± 3
Terbufos	5.0×10^{-6}	41 ± 4	8.9×10^{-6}	23 ± 2
Quinalphos	6.0×10^{-6}	37 ± 11	4.1×10^{-6}	47 ± 10
Dichlorfenthion	8.0×10^{-6}	24 ± 1	6.0×10^{-6}	32 ± 2
Parathion	8.7×10^{-6}	23 ± 2	2.8×10^{-6}	40 ± 8
Mecarbam	9.4×10^{-6}	22 ± 6	8.8×10^{-6}	23 ± 1
Tolclofos-methyl	9.8×10^{-6}	20 ± 1	N.D.	12 ± 1
Leptophos	1.0×10^{-5}	20 ± 5	7.2×10^{-6}	29 ± 2
Pirimiphos-methyl	N.D.	14 ± 7	3.7×10^{-6}	38 ± 3
Disulfoton	N.D.	18 ± 4	7.9×10^{-6}	27 ± 7
Ethoprophos	N.D.	15 ± 1	1.0×10^{-5}	20 ± 3
4. Pyrethroids				
Flucythrinate	7.9×10^{-7}	100 ± 6	7.1×10^{-7}	72 ± 1
Cyhalothrin	8.2×10^{-7}	97 ± 10	1.4×10^{-6}	72 ± 9
Pyrethrin	9.7×10^{-7}	119 ± 12	8.4×10^{-7}	90 ± 2
Deltamethrin	1.1×10^{-6}	98 ± 8	1.4×10^{-6}	68 ± 6
Cyfluthrin	1.4×10^{-6}	88 ± 6	1.9×10^{-6}	49 ± 1
Tralomethrin	1.8×10^{-6}	107 ± 9	9.1×10^{-7}	85 ± 8
Permethrin	1.8×10^{-6}	85 ± 5	2.5×10^{-6}	74 ± 5
Fenvalerate	2.1×10^{-6}	107 ± 4	9.1×10^{-7}	112 ± 3
Fluvalinate	2.1×10^{-6}	59 ± 6	1.1×10^{-6}	138 ± 18

Table 2 (*Continued*)

Compound	hPXR assay		mPXR assay	
	REC_{20}[a]	RLA[b]	REC_{20}[a]	RLA[b]
	(M)	(%)	(M)	(%)
Cypermethrin	2.1×10^{-6}	71 ± 9	2.4×10^{-6}	68 ± 8
Tefluthrin	2.3×10^{-6}	74 ± 6	2.1×10^{-6}	83 ± 3
Etofenprox	2.4×10^{-6}	53 ± 9	2.8×10^{-6}	53 ± 7
5. Carbamates				
Pyributicarb	5.0×10^{-8}	147 ± 7	2.5×10^{-6}	61 ± 7
Esprocarb	6.7×10^{-7}	140 ± 9	5.8×10^{-6}	31 ± 1
Thiobencarb	1.0×10^{-6}	81 ± 4	1.3×10^{-6}	39 ± 4
Dimepiperate	1.9×10^{-6}	73 ± 6	5.4×10^{-6}	40 ± 1
Diethofencarb	6.0×10^{-6}	34 ± 4	N.D.	3 ± 1
Chlorpropham	6.5×10^{-6}	34 ± 1	N.D.	2 ± 1
6. Acid amides				
Pretilachlor	1.0×10^{-7}	68 ± 6	2.5×10^{-6}	30 ± 6
Metolachlor	5.0×10^{-7}	81 ± 6	2.7×10^{-6}	32 ± 4
Thenylchlor	5.8×10^{-7}	66 ± 5	3.2×10^{-6}	25 ± 1
Flutolanil	1.1×10^{-6}	70 ± 9	1.8×10^{-6}	54 ± 7
Alachlor	1.1×10^{-6}	67 ± 2	3.5×10^{-6}	39 ± 1
Mepronil	1.4×10^{-6}	52 ± 5	1.0×10^{-6}	55 ± 10
Mefenacet	2.1×10^{-6}	53 ± 7	6.3×10^{-6}	33 ± 11
Propyzamide	4.9×10^{-6}	40 ± 2	3.9×10^{-6}	40 ± 5
Metalaxyl	8.5×10^{-6}	43 ± 8	N.D.	11 ± 1
Cafenstrole	N.D.	1 ± 2	4.1×10^{-6}	41 ± 6
7. Triazines				
Prometryn	4.9×10^{-6}	32 ± 6	N.D.	6 ± 1
8. Ureas				
Dymuron	4.4×10^{-7}	103 ± 13	N.D.	7 ± 1
Pencycuron	1.4×10^{-6}	78 ± 10	N.D.	9 ± 3
Prochloraz	2.0×10^{-6}	58 ± 7	2.3×10^{-7}	87 ± 2
Linuron	4.3×10^{-6}	60 ± 11	5.2×10^{-6}	36 ± 5
Diuron	8.7×10^{-6}	24 ± 1	N.D.	7 ± 1
9. Others				
Triadimefon	5.4×10^{-7}	96 ± 7	8.5×10^{-6}	23 ± 2
Bromopropylate	6.5×10^{-7}	128 ± 7	2.8×10^{-6}	30 ± 1
Indanofan	7.7×10^{-7}	82 ± 14	1.5×10^{-6}	46 ± 5
Pendimethalin	1.0×10^{-6}	71 ± 1	3.7×10^{-7}	95 ± 13
Trifluralin	1.4×10^{-6}	72 ± 1	3.7×10^{-6}	29 ± 1
Ethoxyquin	1.6×10^{-6}	96 ± 1	4.5×10^{-6}	38 ± 5
Pyrazoxyfen	1.8×10^{-6}	70 ± 12	2.5×10^{-6}	34 ± 14
Fenarimol	2.4×10^{-6}	73 ± 1	3.1×10^{-6}	40 ± 4
Imazalil	2.7×10^{-6}	49 ± 4	1.5×10^{-6}	42 ± 2
Bitertanol	3.6×10^{-6}	44 ± 1	5.6×10^{-6}	29 ± 1
Isoprothiolane.	4.2×10^{-6}	49 ± 4	7.3×10^{-6}	23 ± 4
Ferimzone	4.4×10^{-6}	46 ± 2	N.D.	16 ± 3
Propiconazole	4.7×10^{-6}	36 ± 7	1.1×10^{-6}	68 ± 5
Benfuresate	4.7×10^{-6}	36 ± 2	N.D.	11 ± 3
2,4-PA	6.1×10^{-6}	32 ± 1	2.4×10^{-6}	52 ± 3
Triforine	7.3×10^{-6}	25 ± 1	5.0×10^{-6}	34 ± 2
Procymidone	9.0×10^{-6}	26 ± 1	N.D.	7 ± 1
2-Phenylphenol	9.8×10^{-6}	21 ± 4	N.D.	7 ± 1
Triflumizole	9.9×10^{-6}	20 ± 2	1.5×10^{-6}	38 ± 1

[a] 20% relative effective concentration; the concentration of the test compound showing 20% of the agonistic activity of 1×10^{-5} M rifampicin via hPXR, or 1×10^{-5} M PCN via mPXR. Each REC_{20} value represents the mean of three independent experiments.

[b] Relative luciferase activity; percentage response at a concentration of 1×10^{-5} M with 100% activity defined as the activity achieved with 1×10^{-5} M rifampicin or 1×10^{-5} M PCN. Each RLA value is expressed as mean ± SD from at least three independent experiments performed in triplicate.

[c] RLA of rifampicin for hPXR is represented as the activity at a concentration of 1×10^{-5} M.

[d] Not detectable (no effect or $REC_{20} > 1 \times 10^{-5}$ M).

[e] RLA of PCN for mPXR is represented as the activity at a concentration of 1×10^{-5} M.

PXR was activated by compounds from to all nine groups classified according to their chemical structure. However, clear differences were found in the ratio of the PXR-activating pesticides over all pesticides in each group. The order of the activation ratios was as follows: pyrethroids (100%; 12 of 12) > acid amides (83%; 10 of 12) ≈ organochlorines (83%; 24 of 29) > ureas (71%; 5 of 7) > diphenyl ethers (55%; 6 of 11) > organophosphorus (50%; 28 of 56) > others (43%; 19 of 44) > carbamates (27%; 6 of 22) > triazines (14%; 1 of 7).

Based on their β-galactocidase activity, none of the 200 pesticides in this study showed any cytotoxic effects at the screening doses tested. In addition, we confirmed that the induction of luciferase transcription activity by these pesticides was PXR-dependent as none of the pesticides tested had any effect on the transcriptional activity in the reporter gene assay using COS-7 cells transfected with the reporter plasmid and control plasmid (data not shown).

3.3. Potent PXR activators among the 200 pesticides

Figure 2A and B shows the dose-dependency of hPXR- and mPXR-mediated transcriptional activity induced by four potent hPXR activators (pyributicarb, piperophos, pretilachlor and butamifos) and four potent mPXR activators (phosalone, prochloraz, pendimethalin, and butamifos), respectively. Among these pesticides, pyributicarb and phosalone induced luciferase transcription

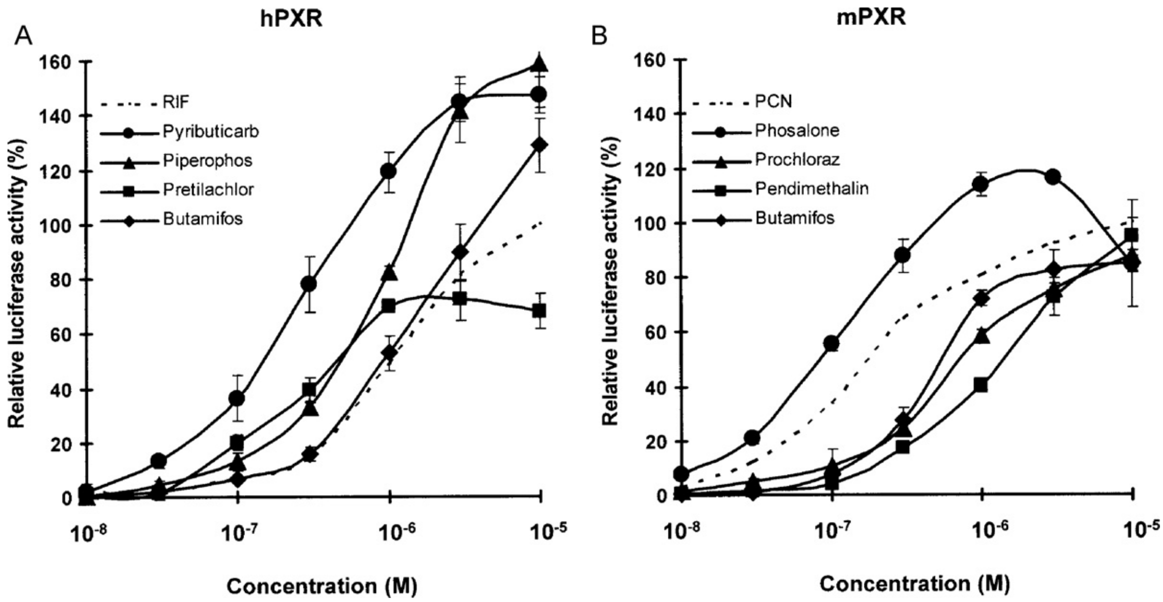

Fig. 2. Dose–response curves of the pesticides showing potent agonistic activity in the hPXR and mPXR transactivation assays. COS-7 cells were transiently transfected with an expression plasmid for hPXR (A) or mPXR (B) as well as a reporter-responsive *firefly* luciferase plasmid and a constitutively active β-galactosidase expression plasmid. The *firefly* luciferase activity was normalized against β-galactosidase activity. Values represent the means ± SD of three independent experiments and are presented as the relative response (%) against that achieved with 1×10^{-5} M RIF (A) and PCN (B) (100%).

via hPXR and mPXR, respectively, at low concentrations in the order of 10^{-8} M. The relative potency of pyributicarb for hPXR was 8.6-fold that of RIF, and the relative potency of phosalone for mPXR was 2.0-fold that of PCN from a comparison of the REC20 values (Table 2). However, pyributicarb and phosalone were found to have species-specific differences in PXR agonistic activity between humans and mice as the REC_{20} values of pyributicarb for mPXR and phosalone for hPXR were in the order of 10^{-6} M (Table 2).

Compared to each RLA at 1×10^{-5} M, 8 pesticides (γ-BHC, chloropropylate, butamifos, isofenphos, piperophos, esprocarb, pyributicarb, and bromopropylate) induced hPXR-mediated gene transcription by more than 1.2-fold greater than did RIF (Table 2). A pyrethroid insecticide, fluvalinate, only induced mPXR-mediated gene transcription by more than 1.2-fold greater than PCN did (Table 2).

Figure 3 describes the chemical structures of several potent hPXR and/or mPXR agonistic pesticides, that showed PXR activation at low concentrations in the order of 10^{-8}–10^{-7} M, together those of with RIF and PCN.

3.4. Comparison of PXR agonistic activity between parent pesticides and their oxon-metabolites

In addition to 200 pesticides, we also examined the PXR activity of the oxon-metabolites of six organophosphorus insecticides (piperophos, butamifos, isofenphos, EPN, isoxathion, and chlorpyrifos), and a carbamate-type herbicide pyributicarb. Figure 4A and B show the dose–responses of the parent pesticides and their oxon-metabolites in the hPXR and mPXR assays, respectively. All six organophosphorus insecticides and pyributicarb showed more potent PXR activity than did their oxon-metabolites in the both of the hPXR and mPXR assays. Table 3 summarizes the REC_{20} values of these pesticides and their oxon-metabolites, as well as relative potency of each oxon-metabolite to the parent pesticide. Although

Table 3
Comparison of PXR agonistic activity between pesticides and their oxon-metabolites.

Compound	hPXR assay		mPXR assay	
	REC_{20}[a] (M)	Relative potency [b]	REC_{20}[a] (M)	Relative potency[b]
Piperophos	1.7×10^{-7}	1	1.0×10^{-6}	1
Piperophos oxon	5.3×10^{-7}	0.32	4.5×10^{-6}	0.22
Butamifos	3.7×10^{-7}	1	2.3×10^{-7}	1
Butamifos oxon	1.7×10^{-6}	0.22	8.5×10^{-7}	0.27
Isofenphos	6.4×10^{-7}	1	6.8×10^{-7}	1
Isofenphos oxon	1.7×10^{-6}	0.38	1.9×10^{-6}	0.36
EPN	1.8×10^{-6}	1	1.7×10^{-6}	1
EPN oxon	N.D. [c]	<0.18	N.D.	<0.17
Isoxathion	2.0×10^{-6}	1	5.1×10^{-6}	1
Isoxathion oxon	N.D.	<0.20	N.D.	<0.51
Chlorpyrifos	4.7×10^{-6}	1	4.5×10^{-6}	1
Chlorpyrifos oxon	N.D.	<0.47	N.D.	<0.45
Pyributicarb	5.0×10^{-8}	1	2.5×10^{-6}	1
Pyributicarb oxon	4.9×10^{-7}	0.11	5.0×10^{-6}	0.5

[a] 20% relative effective concentration; the concentration of the test compound showing 20% of the agonistic activity of 1×10^{-5} M RIF via hPXR, or 1×10^{-5} M PCN via mPXR.
[b] Relative potency is expressed as a ratio of REC_{20} of the parent pesticide for hPXR or for mPXR to that of the oxon-metabolite; i.e., the relative potency of each parent pesticide for hPXR or mPXR is arbitrarily set at 1.
[c] Not detectable (no effect or $REC_{20} > 1 \times 10^{-5}$ M).

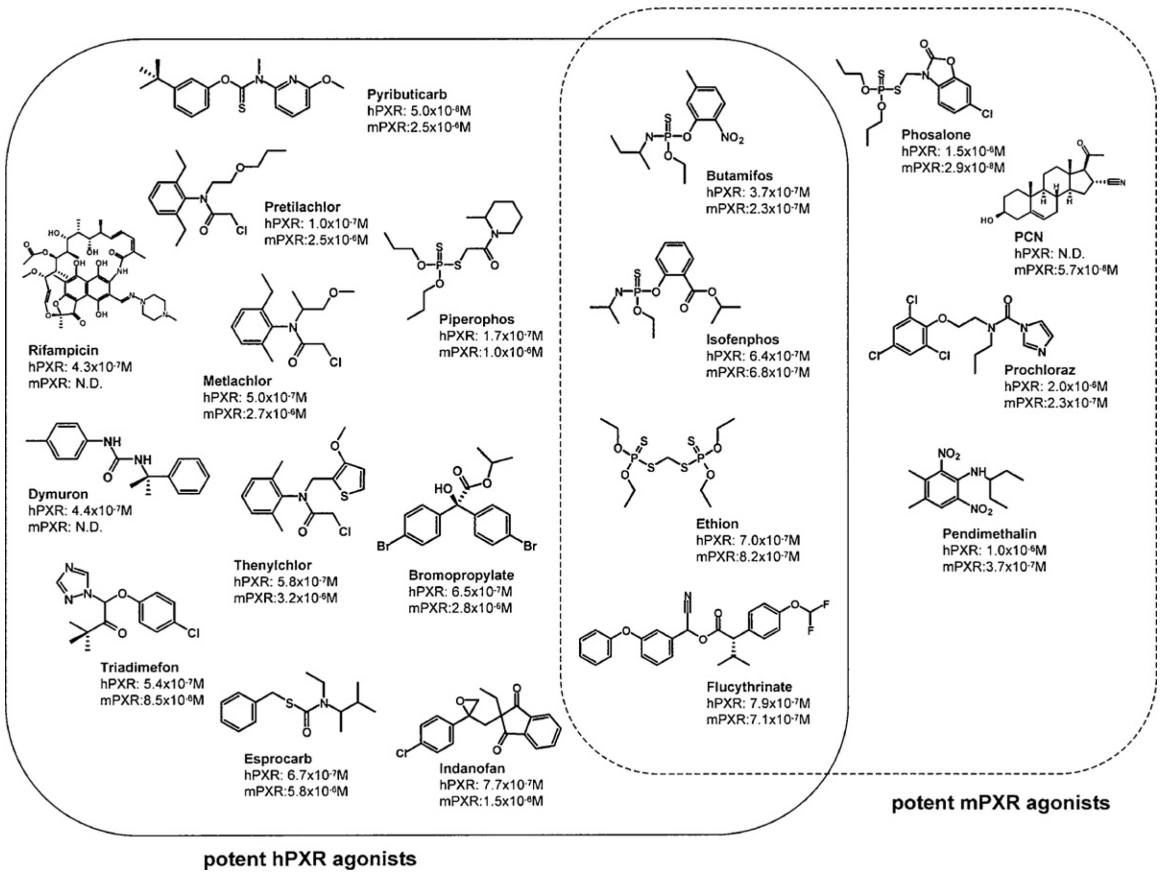

Fig. 3. Chemical structures of PXR ligands and several pesticides showing potent hPXR and mPXR agonistic activity. The chemicals within the solid and dashed lines represent potent hPXR and mPXR agonists, respectively. Values represent the REC_{20} for hPXR and mPXR.

three oxon-metabolites of EPN, isoxathion, and chlorpyrifos show activities of more than 20% below of the maximal activity of RIF and PCN, a comparison of the REC_{20} values shows that the other four oxon-metabolites activated hPXR and mPXR at 2.7–9.8 fold and 2.1–4.4 fold higher concentrations than did their parent pesticides, respectively.

4. Discussion

PXR is an important regulator of several steroid and xenobiotic detoxification enzymes (e.g., CYP3A, CYP2B, and UGT1A), as well as transporters in the liver and intestines (Kretschmer and Baldwin, 2005; Timsit and Negishi, 2007). Since PXR activation by various chemicals unexpectedly induces CYP3A and CYP2B, leading to alterations in the metabolism of critical developmental factors such as steroids or other endogenous molecules, it has been speculated that PXR agonists have the potential to act as endocrine disruptors (Tabb and Blumberg, 2006). To date, PXR agonistic activity has been found in a number of xenobiotics, such as drugs and environmental contaminants including pesticides (Martin et al., 2010; Zhou et al., 2009). However, despite the LBD of PXR displaying species differences between humans and rodents, there are few reports comparing human and rodent PXR activation by numerous pesticides. This may result in misinterpretations in health risk assessments of pesticides for humans when based on animal experiments using rodents.

In this report, we show the responsiveness of hPXR and mPXR to 200 pesticides, all of which had been already screened for ERα, ERβ, AR, PPARα, PPARγ and AhR activities (Kojima et al., 2004; Takeuchi et al., 2006, 2008). Surprisingly, we have found that 106 and 93 of the 200 pesticides, respectively, possessed hPXR and mPXR agonistic activity in COS-7 cell-based reporter gene assays, including many compounds newly identified as PXR activators in this study. A total of 111 pesticides had agonistic activity against hPXR and/or mPXR, while the level of activity showed greater or lesser species differences (Table 2). Thus, our findings confirm that PXR is a promiscuous nuclear receptor that responds to structurally diverse ligands, and acts physiologically as a xenobiotic sensor (Kliewer et al., 2002; Kretschmer and Baldwin, 2005). In addition to the activation of PXR described here, a large number of pesticides also had multiple effects on transcriptional activity via other nuclear hormone receptors (Kojima et al., 2004), leading to potential effects on both the endocrine system and reproduction in whole organisms.

Our study demonstrated that 24 of 29 organochlorines, including their isomers, had PXR agonistic activity, with the BHC isomers, chloropropylate and methoxychlor in the hPXR assay, and dicofol and methoxychlor in the mPXR assay, showing particularly potent activity (Table 2). Coumoul et al. (2002) reported that the organochlorines, chlordane, α-endosulfan, and dieldrin, acted as hPXR agonists in transactivation assays using HepG2 and human breast carcinoma MCF-7 cells, and α-endosulfan could, additionally, increase CYP3A4 mRNA levels in human primary hepatocytes. Using the stable hPXR/HepG2 cell line, Lemaire et al. (2004) showed that the order of hPXR agonistic activity of organochlorines was as follows: γ-BHC > endosulfan > methoxychlor, chlordane > dieldrin > DDT > aldrin > PCP. Thus, our results support those of the cited reports (Coumoul et al., 2002; Lemaire et al., 2004), and also show that mPXR is activated by various organochlorine pesticides at a concentration range of 10^{-6} M.

We newly identified six diphenyl ether-type herbicides as having both hPXR and mPXR agonistic activities. Recently, it has been

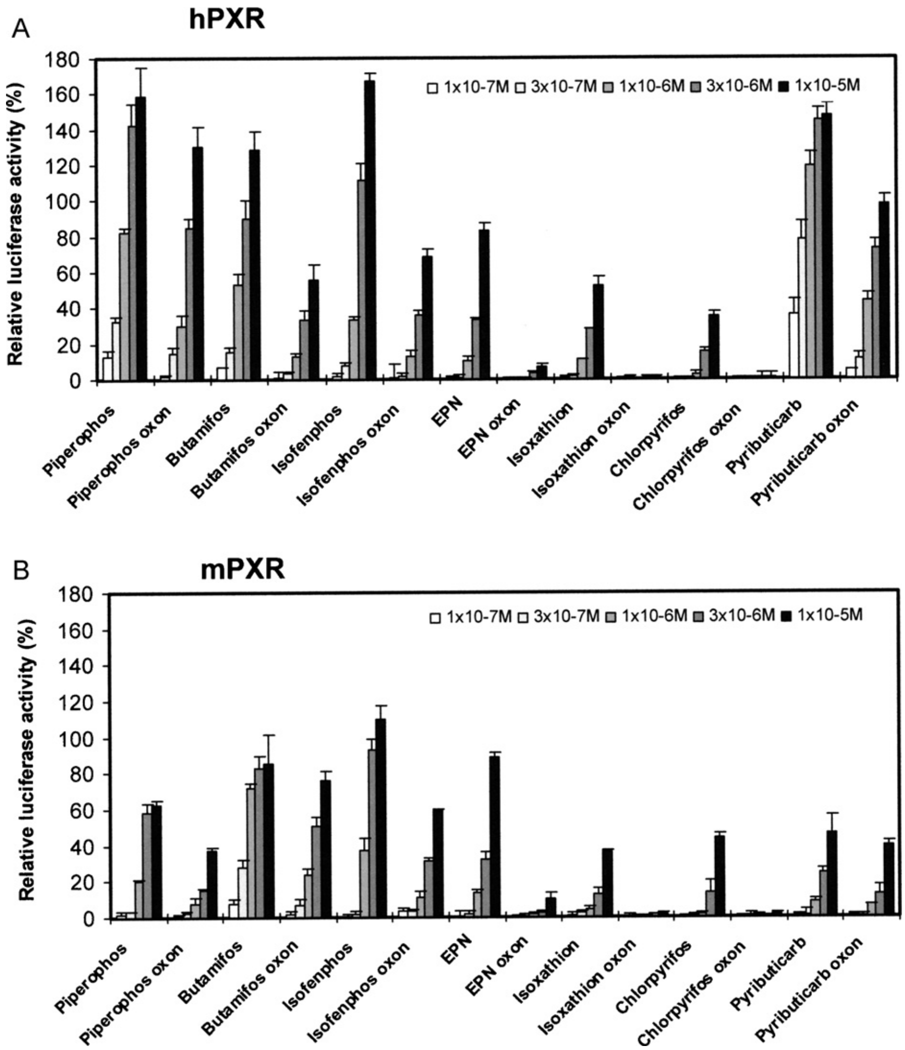

Fig. 4. Dose–responses of pesticides and their oxon-metabolites on the hPXR- and mPXR-mediated transcriptional activities. COS-7 cells were transiently transfected with an expression plasmid for hPXR (A) or mPXR (B) as well as a reporter-responsive *firefly* luciferase plasmid and a constitutively active β-galactosidase expression plasmid. Cells were treated with 1×10^{-7} to 1×10^{-5} M of the six organophosphorus insecticides, pyributicarb, and their oxon-metabolites. The *firefly* luciferase activity was normalized against β-galactosidase activity. Values represent the means ± SD of three independent experiments and are presented as the relative response (%) against that achieved with 1×10^{-5} M RIF (A) and PCN (B) (100%).

reported that an antimicrobial 2′-hydroxyl 2,4,4′-trichlorinated diphenyl ether (triclosan) and polybrominated diphenyl ethers (PBDEs), used as flame retardants, showed hPXR agonistic activity (Jacobs et al., 2005; Pacyniak et al., 2007). Therefore, these findings suggest that halogenated compounds having a diphenyl ether structure might act as PXR agonists. Interestingly, we previously showed that these six diphenyl ether-type herbicides, and several PBDEs, including their metabolites, were potent AR antagonists (Kojima et al., 2003, 2004, 2009).

To our knowledge, there have been few reports on the PXR activation of organophosphorus pesticides apart from evidence showing that chlorpyrifos, fenitrothion, EPN, and isoxathion significantly increase reporter activity via hPXR in HepG2-derived cells stably expressing the *CYP3A4* promoter/enhancer (Matsubara et al., 2007). We tested 56 organophosphorus pesticides, and consequently found that 28 pesticides, including chlorpyrifos, EPN and isoxathion, possessed hPXR and/or mPXR agonistic activity. These pesticides included several potent PXR agonists that stimulate the transcription of luciferase at very low concentrations in the order of 10^{-8}–10^{-7} M. For example, piperophos, phosalone and butamifos activated hPXR and/or mPXR at concentrations similar to or lower than those of RIF or PCN (Fig. 2A and B). Interestingly, organophosphorus insecticides activating PXR appear to present some specific structural requirements that display no species difference; e.g., pesticides having a diethylthiophosphoryl residue ($(CH_3CH_2O)_2$–P(=S)–), such as chlorpyrifos and bromophos-ethyl, are more potent PXR activators than those having a dimethylthiophosphoryl residue ($(CH_3O)_2$–P(=S)–), such as chlorpyrifos-methyl and bromophos-methyl. The relationship between chemical structure and PXR activation was confirmed in other organophosphorus pesticides tested in this study. Furthermore, as shown in Fig. 4 and Table 3, organophosphorus pesticides having a thiophosphoryl (P=S) residue, such as butamifos and EPN, are more potent PXR activators than those having an oxyphosphoryl (P=O) residue, such as butamifos-oxon and EPN-oxon. These decreases in receptor activity observed in the oxon-forms could be also shown in other receptors of ER, AR, and AhR (Kojima et al., 2004; Takeuchi et al., 2008). Therefore, a thiophosphoryl residue may be positively involved in interactions with nuclear receptors and AhR, suggesting that organophosphorus insecticides having this residue would show decreased receptor activities, including PXR activation, on changing the thiophosphoryl residue to an oxyphosphoryl residue during oxidative metabolization in the environment or body.

We revealed that all of the 12 pesticides belonging to the pyrethroid group showed both hPXR and mPXR agonistic activities at a concentration of 10^{-7}–10^{-6} M without any species differences between humans and mice. Prior studies reported that fenvalerate, cypermethrin and permethrin are potent activators of hPXR (Lemaire et al., 2004, 2006) and 10 pyrethroids, apart from tetramethrin, significantly induce CYP3A4 via PXR (Yang et al., 2009). Together with these reports, our data strongly indicate that pyrethroids are the most likely suspected candidates for PXR activators among the various types of pesticides.

Of the 22 carbamate-type pesticides tested, only six showed hPXR and/or mPXR agonistic activities. However, one carbamate-type herbicide, pyributicarb, showed the most potent hPXR agonistic activity among all the 200 pesticides tested. The REC_{20} value of this herbicide for hPXR was 5.0×10^{-8} M, and was 8.6-fold stronger than that of RIF, although its REC_{20} value for mPXR was only in the order of 10^{-6} M (Table 2 and Fig. 3). Furthermore, we found that the hPXR agonistic activity of pyributicarb oxon-metabolite was 9.8-fold weaker than that of pyributicarb, and that the oxon-metabolites of organophosphorus pesticides also showed weaker PXR activity than did their parent compounds (Table 3). However, this oxon-metabolite showed hPXR activity similar to that of RIF. Previously, Matsubara et al. (2007) reported that, among the 18 pesticides tested, pyributicarb was a particularly potent activator of the CYP3A4 gene, and its hPXR activation was also confirmed in an *in vivo* model using transgenic mice with the hPXR gene. This *in vivo* data may also support our evidence that pyributicarb and its oxon-metabolite have very potent hPXR agonistic activity.

Previous reports have shown that five acid amide-type pesticides, pretilachlor, metlachlor, alachlor, mepronil and flutolanil, are all hPXR activators (Lemaire et al., 2006; Matsubara et al., 2007), and that alachlor is a rat PXR activator (Mikamo et al., 2003). Our study showed that 10 of 12 acid amide-type pesticides, including the above five PXR activators, possessed hPXR and/or mPXR activity. Among these compounds, pretilachlor, metlachlor, thenylchlor and alachlor have a similar structure, and show more potent hPXR than mPXR agonistic activities (Fig. 3).

Among the seven triazine-type pesticides tested, only prometryn showed weak agonistic activity against hPXR, and none of the triazines activated mPXR. This type of pesticide appears to be structurally inactive with regard to PXR activation. While urea-type pesticides are mainly used as herbicides, our study indicated that five of seven urea-type herbicides act as PXR activators, and daimuron and prochloraz, in particular, have potent hPXR and mPXR agonistic activity, respectively (Fig. 3). In addition to PXR activity, linuron and diuron have been shown to be AR antagonists and AhR agonists (Kojima et al., 2004; Takeuchi et al., 2008).

In the present study, 44 pesticides that could not be classified into the above eight groups were collectively grouped as "others". Among these pesticides, vinclozolin was shown to be a rat PXR activator (Mikamo et al., 2003); however, we did not observe its PXR activation in our study using hPXR and mPXR assays, nor was its hPXR activation reported by Lemaire et al. (2006). This discrepancy may reflect the fact that the PXR activation by vinclozolin has marked species differences, not only between humans and rodents, but also between mice and rats. Our study using pesticides classified as "others" showed that triadimefon, bromopropylate and indanofan preferentially activated hPXR, whereas pendimethalin preferentially activated mPXR (Fig. 3). Thus, unlike other nuclear receptors, PXR exhibits considerable differences in its activation among mammals.

Our study revealed that 111 of 200 pesticides tested have hPXR and/or mPXR agonistic activity, and might potentially induce CYP3A and other target genes involved in various important physiological reactions such as drug, steroid and glucose metabolism. Most recently, Martin et al. (2010) has reported that 234 and 102 of 320 pesticides activated the PXRE via endogenous receptors in HepG2 cells (CIS assay) and the GAL4-PXR ligand-binding domain exogenously expressed in HepG2 cells (TRANS assay), respectively. Seventy-four pesticides used in these assays appear to overlap with those selected in our study. Although the results of CIS and TRANS assays for these pesticides well coincided with the data from our study of hPXR, there are some differences between the two data sets. This discrepancy may reflect the fact that we used the exogenous full-length protein of PXR and the host COS-7 cells rather than HepG2 cells. Among environmental contaminants apart from pesticides, bisphenol-A, nonylphenol and phthalates have been reported to be PXR agonists as well as endocrine disruptors via ER and AR (Masuyama et al., 2000; Mnif et al., 2007; Takeshita et al., 2001). Furthermore, recent studies showed that PXR agonistic activity could be detected in environmental wastewater samples using an hPXR reporter gene assay (Creusot et al., 2010; Mnif et al., 2010). Thus, in addition to PXR being also activated by a variety of compounds in the environment, our findings suggest that the number of its ligands might be significantly larger than the number of ligands against nuclear hormone receptors such as ER and AR. This might support that PXR can bind structurally diverse ligands due to the large size ($\geq$1200 Å^3) of its ligand-binding domain relative to other nuclear receptors (Watkins et al., 2001). Taken together, our results demonstrate that numerous environmental chemicals can unexpectedly affect the PXR-mediated signaling pathways in humans and rodents, and, therefore, a mixture of these chemicals may show dose-additive effects and cause adverse effects even when each compound is present at a low concentration.

5. Conclusions

In this study, we characterized the hPXR and mPXR agonistic activity of 200 pesticides, and revealed that a large number of pesticides and their metabolites possess these activities with greater or lesser inter-species differences. These PXR activators included, not only organochlorines and pyrethroids, but representatives from all groups studied. In particular, several potent PXR activators, such as pyributicarb and phosalone, showed distinct species-differences between humans and mice. Our data on the relationship between PXR activities, chemical structures and species differences are very useful for predicting the harmful effects of pesticides as well as for developing new pesticides. Although PXR regulates the expression of Phase I and Phase II drug-metabolizing enzymes and transporters, recent studies suggest that PXR is also involved in energy metabolism through its effects on the metabolism of fatty acids, lipids and glucose (Wada et al., 2009). Most recently, it has been shown that RIF exhibits an inhibitory effect on γ-interferon production via PXR in human T lymphocytes (Dubrac et al., 2010), raising a concern that unexpected exposure to PXR activators may change the balance of ongoing immune responses. Thus, further studies, such as the screening of environmental contaminants for PXR, are required to predict the potential adverse effects of these chemicals on energy metabolism, and the endocrine and immune systems.

Conflict of interest

None.

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